Morphogenic Intelligence: Runtime Neural Development Beyond Static Architectures*

Lisa Welsch
lisa@tastehub.io

Martyna Kwiecień
martyna@tastehub.io

April 2026

Abstract

We introduce *Morphogenic Intelligence* (MI), a neural architecture in which compute units — called *Morphons* — grow, differentiate, fuse, and undergo apoptosis at runtime. Unlike Transformers, CNNs, RNNs, and SSMs, where topology is fixed at design time and only weights are learned, MI treats network architecture itself as a primary learned artifact. Each Morphon carries a developmental program inspired by biological morphogenesis: chemical gradients in a hyperbolic embedding space guide connectivity, neuromodulatory signals gate plasticity, and a dual-clock scheduler separates fast inference from slow structural remodeling. We implement the full MI engine in Rust and evaluate it on CartPole control, MNIST classification, and a synthetic NLP readiness benchmark. CartPole-v1 is solved (avg=195.2 on the v0.5.0 codebase, demonstrated learning at avg=166 on the current v3.0.0 quick profile) using only local three-factor learning and developmental morphogenesis. On MNIST, the system reaches 30% intact accuracy and **48%** after destroying 30% of associative neurons and allowing the network to regrow — a self-healing response that exceeds the intact baseline by 18 percentage points. We document and fix four characteristic failure modes (modulatory explosion, motor silencing, LTD vicious cycle, premature regulatory throttling), each of which has a direct biological analogue. The premature throttling bug — where the homeostatic regulator declares a system "mature" at 11% accuracy because dense supervised reward inflates its slow EMA — is novel and broadly applicable to any biologically-inspired regulator that uses raw reward as a maturity signal. While MI does not yet match the classification accuracy of mature static architectures, it demonstrates qualitatively distinct capabilities: zero-shot structural adaptation, graceful degradation under ablation, and the surprising property of *improving* after damage.

1 Introduction

All dominant neural architectures — Transformers [24], CNNs [14], RNNs, and state-space models [10] — share a common assumption: *topology is fixed before training begins*, and learning is reduced to weight optimization within that frozen graph. Neural architecture search and dynamic pruning are offline approximations to this constraint, not runtime alternatives to it. This design choice has yielded remarkable results, but it also incurs costs that are increasingly visible as systems scale: catastrophic forgetting, the need for massive overparameterization to compensate for lack of structural adaptation, and the inability to allocate capacity where it is needed at runtime.

Biological neural systems do not work this way. Cortical circuits develop, prune, and reorganize continuously throughout life [11, 19]. Adult neurogenesis persists in the hippocampus and olfactory bulb [4, 15]. Synaptic boutons form and disappear on timescales of hours [21]. This structural plasticity is not incidental to biological intelligence — it is computationally essential. The cortex does not merely learn weights; it grows the substrate of computation in response to its inputs.

We argue that architectures capable of growing their own topology under local developmental rules will exhibit qualitatively different generalization and adaptation properties than static networks. We call this paradigm *Morphogenic Intelligence* (MI), and we present here the first complete implementation: a Rust-based simulator in which each compute unit (a *Morphon*) carries a developmental program, communicates with neighbors through a biophysically-inspired resonance mechanism, and is born, differentiates, fuses, or dies according to local rules and metabolic budgets.

*Preprint. DOI: 10.5281/zenodo.19467243. Code: https://github.com/SimplyLiz/Morphon

Contributions.

1. A complete developmental neural architecture grounded in six biological principles (Section 3).
2. A Rust implementation of the full MI engine, including all six principles, that runs in real time on commodity hardware (Section 4).
3. Experimental validation on CartPole (solved), MNIST (30% intact, 48% post-damage on v3.0.0), and a 4-tier NLP readiness benchmark (Section 5).
4. A taxonomy of biology-informed failure modes encountered during development, each with a biological analogue and a biology-derived fix that transfers to the computational setting (Section 6).
5. A novel regulatory bug — premature "Mature" stage detection in dense-reward classification settings — that affects any biologically-inspired homeostatic system using raw reward as a maturity signal, and a clean fix.

The rest of the paper is organized as follows. Section 2 positions MI relative to prior self-modifying architectures. Section 3 formalizes the six biological principles and the data structures that implement them. Section 4 describes the Rust engine. Section 5 presents the benchmark results. Section 6 is the analytical core of the paper: four failure modes, their biological parallels, and the fixes that resolve them. Section 7 discusses limitations and the path to language tasks. Section 8 concludes.

2 Related Work

We position MI within the landscape of biologically-inspired and self-modifying neural architectures along three axes: (i) whether topology changes at runtime, (ii) whether the developmental rules are local, and (iii) whether the system integrates neuromodulation with structural plasticity.

Topology growth via evolution: NEAT and descendants. NEAT [20] introduced the now-canonical pattern of evolving neural network topologies through speciation and historical-marking-based crossover. NEAT and its descendants grow topology through an offline evolutionary outer loop — generations of mutation and selection — rather than through runtime developmental rules. The key difference from MI: NEAT topology change happens between training runs (each genome is fixed once instantiated), while MI topology changes during a single run under local energy and activity constraints. NEAT also has no apoptosis, no neuromodulatory gating, and no structural memory hierarchy.

Neural Developmental Programs (NDP). NDP [16] learns local developmental rules that produce a network topology, treating the developmental program itself as the optimization target. NDP uses differentiable relaxations and gradient-based meta-learning of the developmental rules. MI is the dual approach: the developmental rules are hand-specified (grounded in the six biological principles of Section 3) and run during inference, with discrete morphogenetic events (birth, fusion, apoptosis) gated by metabolic costs. NDP learns to grow networks; MI grows networks while learning.

Three-factor learning rules. The three-factor framework [5, 8] extends Hebbian plasticity with a neuromodulatory third factor (typically a dopamine analogue gating eligibility-traced LTP). MI implements three-factor learning with receptor-gated modulation: each Morphon's response to each modulator is determined by its evolving receptor density profile, allowing morphon-specific sensitivity to develop without a global hyperparameter.

Spiking neural networks for MNIST: Diehl & Cook. The closest comparison point for our MNIST experiments is Diehl & Cook [3], which achieves ∼95% test accuracy on MNIST using unsupervised STDP with adaptive thresholds and lateral inhibition. Our MNIST setup explicitly draws on their two-phase pattern (unsupervised feature formation followed by post-hoc neuron labeling), and we use their accuracy as the realistic ceiling for STDP-based spiking classification — not the 99%+ regime of fully-supervised SNNs trained with surrogate gradients. Our intact 30% / post-recovery 48% is well below this ceiling and there is room to improve; the paper's contribution is the developmental dynamics and the failure-mode catalogue, not the accuracy itself.

Cortical Labs (DishBrain). DishBrain [13] demonstrated that biological neurons cultured on multi-electrode arrays can learn to play Pong via reward-modulated plasticity. This is empirical evidence that self-organizing wetware solves real-time control without backpropagation. MI is the in-silico analogue: a developmental simulation that exhibits the same intrinsic-activity-plus-reward-modulation principles.

Synaptic scaling and homeostatic regulation. Turrigiano's synaptic scaling [22, 23] and BCM metaplasticity [1] provide the biological grounding for MI's homeostatic mechanisms. Our *Endoquilibrium* module (Section 3.5) implements multi-scale homeostatic regulation and is the locus of the regulatory failure mode we document in Section 6.

Synaptic tagging and capture. Frey & Morris [7] established the tag-and-capture model of late-LTP consolidation: weak inputs leave a transient "tag" that can be converted to long-lasting plasticity if a strong, protein-synthesis-triggering signal arrives within hours. MI implements this directly (Section 4), and our finding that capture must be *episode-gated* rather than per-tick (Section 4, tag-and-capture subsection) is, we believe, a useful refinement for any spiking architecture using this mechanism.

Inhibitory plasticity (iSTDP). Vogels et al. [25] showed that iSTDP on inhibitory synapses balances excitation and inhibition in cortical-like networks. MI's planned local inhibitory competition (Section 7) is a direct application: replace the global k-WTA approximation with trained inhibitory interneurons that learn the right competition pattern through Vogels-style iSTDP.

3 Architecture

This section presents the six biological principles underlying MI and the data structures that implement them. Each subsection maps a biological concept to a precise computational mechanism. Figure 1 shows the overall dual-clock loop and how Endoquilibrium modulates each subsystem.

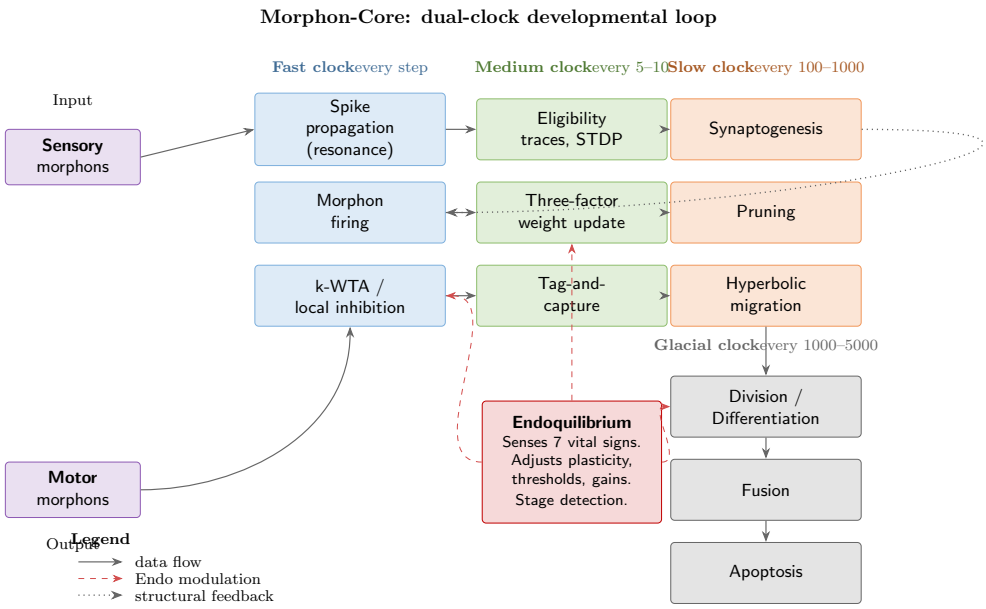

Figure 1: Morphon-Core dual-clock developmental loop. Solid arrows are data flow; dashed red arrows are Endoquilibrium modulation; dotted arrow is the structural-feedback path where slow-clock synaptogenesis introduces new connections that the fast clock then uses. The four temporal scales (fast/medium/slow/glacial) operate on the same morphon population at different periods.

3.1 The Six Biological Principles

P1: Local computation only. No global loss, no backpropagation. All updates use locally available information — pre-synaptic and post-synaptic activity, locally diffused neuromodulators, and a Morphon's own state.

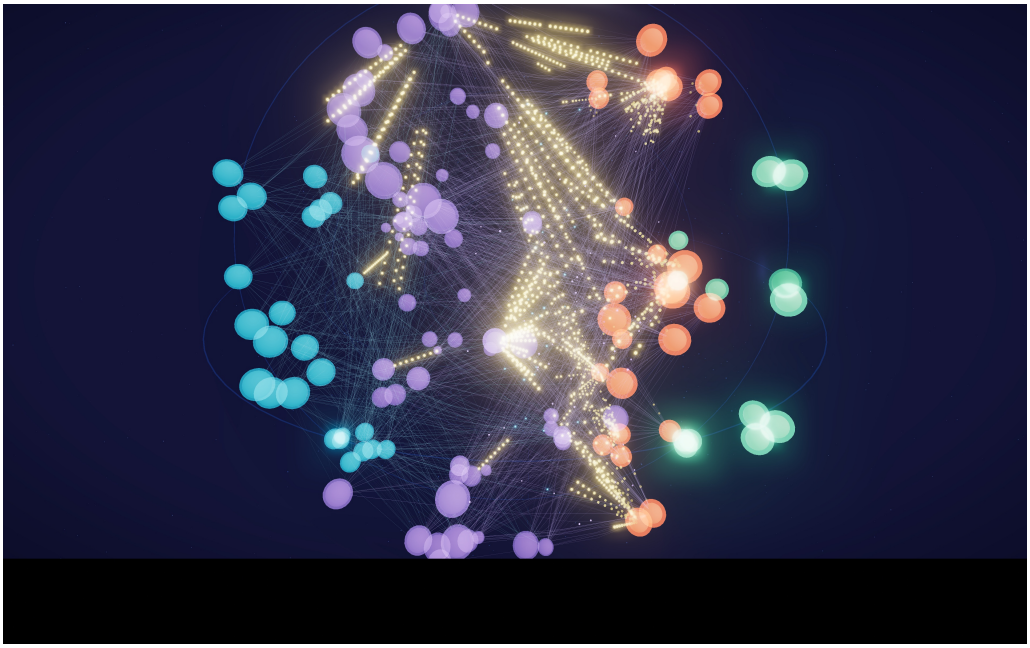

Figure 2: The Morphon network during runtime, rendered by the in-browser WASM visualizer. Spheres are morphons colored by cell type: purple = associative, cyan = sensory, orange = motor, green = modulatory. Dotted yellow trails are spike-carrying synapses; the discrete dots represent action potentials propagating along axons with their learned delays. The 3D layout is the Poincaré ball embedding (Section 3); morphons of the same type cluster naturally because their hyperbolic positions are correlated with cell-type identity. Orange motor morphons concentrate on one side and project to the others through the synaptic web. The visualizer is reproducible from `web/` via `wasm-pack build` and a local HTTP server.

P2: Developmental lifecycle. Morphons are born, differentiate, mature, fuse with correlated neighbors, and can die from energy starvation or chronic inactivity (apoptosis).

P3: Chemical signaling. Morphogen-like signals diffuse through the embedding space and guide differentiation and connectivity decisions.

P4: Neuromodulatory gating. Plasticity rules are gated by four modulatory channels (dopamine, serotonin, acetylcholine, norepinephrine analogues), each with a per-Morphon receptor density.

P5: Multi-scale memory. Synaptic (fast, milliseconds), structural (medium, minutes), and morphogenetic (slow, hours) memory operate on separate timescales.

P6: Metabolic cost. Every Morphon consumes energy each tick. Resource scarcity drives pruning, fusion, and apoptosis.

3.2 The Morphon

The Morphon is the fundamental compute unit. It carries:

- **Membrane state:** potential, prediction-error baseline, refractory timer, fired flag.
- **Receptor profile:** per-modulator sensitivity vector $(\rho_R, \rho_N, \rho_A, \rho_H)$ for reward, novelty, arousal, and homeostasis channels.
- **Developmental metadata:** cell type (Stem, Sensory, Associative, Motor, Modulatory, Fused, Inhibitory Interneuron), differentiation level, lineage ID, birth tick.
- **Energy budget:** current energy, basal regeneration rate, firing cost (with optional superlinear penalty), apoptosis threshold.
- **Position:** hyperbolic coordinate in a Poincaré ball of configurable dimensionality.

A Morphon's state update is local: it integrates spike inputs from incoming synapses, applies an activation function determined by its cell type, fires if the result exceeds its (adaptive) threshold, and pays the metabolic cost.

3.3 Resonance and Spike Propagation

Morphons communicate via discrete spikes propagated along synapses with configurable delays. Unlike Transformer attention (a global $O(N^2)$ operation), spike propagation is local: $O(k \cdot N)$ where k is the average out-degree. The system's cost scales with the number of *active* morphons and their connectivity, not with the square of the population.

Each synapse carries a weight, a delay, an eligibility trace, a tag (for synaptic tagging and capture [7]), and a consolidation level. The dual-clock scheduler updates synaptic weights on the medium path via a three-factor rule:

$$\Delta w = \eta \cdot e \cdot M(t) \tag{1}$$

where e is the eligibility trace built from STDP coincidences, $M(t)$ is the (receptor-gated) modulatory signal, and η is a per-synapse plasticity rate.

3.4 Morphogenesis Lifecycle

The morphogenesis subsystem implements Principles P2 and P3. On the slow clock:

- **Synaptogenesis** forms new synapses between morphons whose activity is correlated and whose hyperbolic distance is below threshold.
- **Pruning** removes synapses with low weight, low recent activity, or whose pre/post morphons are no longer correlated.
- **Migration** moves morphons through hyperbolic space along gradients of correlated input (similar morphons drift together).

On the glacial clock:

- **Division** produces a daughter morphon from a parent that is consistently energetic and active.
- **Differentiation** commits a stem morphon to a specific cell type based on local morphogen concentrations.
- **Fusion** merges two morphons whose activity patterns have become indistinguishable, redistributing their synapses to the survivor.
- **Apoptosis** removes morphons that have been below their energy threshold for a configurable duration.

3.5 Endoquilibrium: Predictive Neuroendocrine Regulation

Endoquilibrium is the homeostatic regulator. It monitors seven vital signs (firing rates by cell type, eligibility density, weight entropy, energy utilization, prediction error mean, reward EMA), maintains dual-timescale exponential moving averages (fast $\tau = 50$, slow $\tau = 500$), and adjusts seventeen modulation channels through proportional control. The seven core channels are receptor gains $\rho_R, \rho_N, \rho_A, \rho_H$ for reward/novelty/arousal/homeostasis, threshold bias, plasticity multiplier, and consolidation gain. Ten additional channels (added incrementally as the V2 primitives shipped) tune structural plasticity parameters: winner adaptation, capture threshold, rollback PE threshold, $\tau_{\text{eligibility}}$, slow trace leak, division threshold, pruning threshold, frustration sensitivity, migration rate, and synaptogenesis threshold.

The most critical role of Endoquilibrium is breaking the firing-rate-zero deadlock. Without it, associative morphons stop firing within ∼50 ticks of training start: low firing rate → no eligibility traces → no weight updates → low firing rate persists. This is a positive-feedback failure that no amount of parameter tuning fixes. Endoquilibrium's Rule 1 (firing rate regulation) detects $\text{FR}_{\text{assoc}} <$ setpoint and lowers thresholds, boosts arousal, and boosts novelty until firing recovers. The fast EMA ($\tau = 50$) responds within ∼50 ticks, just fast enough.

Endoquilibrium also detects developmental stage from the *relative* reward trajectory (not absolute reward levels, which differ wildly between tasks):

Stage	Trigger
Proliferating	history < 20 ticks (warmup)
Stressed	reward trend $< -0.05 \cdot \lvert\text{slow EMA}\rvert$
Mature	reward stable, low cv, history ≥ 2000 ticks
Consolidating	reward near own ceiling, stable, history ≥ 500
Differentiating	reward actively climbing (default healthy state)

The history gates (2000 / 500 ticks) are the result of fixing the *premature Mature* bug discussed in Section 6: dense reward signals on classification tasks inflated the slow EMA before the system had learned anything, triggering Mature at 11% accuracy and locking learning out of the high-plasticity Differentiating regime.

3.6 V2 Developmental Primitives

Three primitives extend the basic morphogenesis loop with additional biological mechanisms. They are individually optional and largely orthogonal to the headline benchmarks, but they shape long-horizon network health.

Frustration-driven exploration. Each morphon maintains a stagnation counter that increments when its prediction error fails to decrease. Above a threshold ($\sim$0.3), the morphon enters a frustration state: noise amplitude increases, weight perturbations are injected, and migration constraints relax. This is a local escape mechanism for morphons stuck in poor configurations and is biologically motivated by stress-driven plasticity.

Bioelectric field. A 2D scalar field over the Poincaré disk holds five named layers (prediction error, energy, stress, novelty, identity). The field diffuses on the slow clock via a discrete heat equation and provides a non-synaptic communication channel: morphons read the local field value and bias migration toward low-PE / high-energy regions. This is the in-silico analogue of neurotrophic gradients in developing cortex.

Target morphology and self-healing. Optional *target regions* specify ideal cell type, density, and position in hyperbolic space. The glacial clock monitors regional underpopulation and either boosts division pressure for nearby morphons or seeds pre-committed morphons in empty regions. This is the mechanism that enables the post-damage recovery in Section 5.2: when 30% of associative morphons are killed, target morphology drives the regrowth that produces the +18 pp self-healing result.

3.7 Hyperbolic Embedding Space

Morphons live in a Poincaré ball model of hyperbolic space with configurable dimensionality (typically 4–6). Hyperbolic geometry has two properties that make it suitable for developmental architectures: (i) tree-like hierarchies are represented with low distortion [17], and (ii) volume grows exponentially with radius, providing room for unbounded topology expansion without coordinate crowding.

In MI, the origin of the ball corresponds to the most general (stem-like) morphons, and the boundary to the most specialized. A morphon's hyperbolic distance to its peers governs synaptogenesis probability and morphogen diffusion.

3.8 Dual-Clock Scheduler

Four temporal scales separate fast inference from slow structural change:

Scale	Period	Operations
Fast	1 step	Spike propagation, firing, input integration
Medium	5–10	STDP, eligibility, three-factor weight updates
Slow	100–1000	Synaptogenesis, pruning, migration
Glacial	1000–5000	Division, differentiation, fusion, apoptosis

This separation is essential. Structural changes during inference would destabilize the network; the slow clock guarantees that morphogenetic events happen between bursts of computation.

4 Implementation

The MI engine is implemented in Rust (∼10,000 lines, 180+ unit tests). The choice of Rust is motivated by three requirements: (i) memory safety without GC pauses, (ii) zero-cost abstractions for the inner simulation loop, and (iii) safe concurrency for parallel morphon updates. The crate compiles to native, WebAssembly (for an interactive in-browser visualizer), and Python bindings via PyO3 (for benchmark harnessing).

4.1 Core Data Structures

Morphons are stored in a `HashMap` keyed by stable IDs. The topology is a directed graph using the `petgraph` crate, with synapses as edge weights. We use generational indices so that references survive birth/death events.

4.2 The Main Loop

Algorithm 1 System::step (one fast tick)

```
1. Spike propagation: deliver pending spikes from previous step
2. Inhibitory competition (k-WTA or local interneurons)
3. Morphon update: integrate inputs, decide firing, pay metabolic cost
4. Generate new spikes from fired morphons
if tick.medium then
    5. Update eligibility traces, apply three-factor learning
    6. Run Endoquilibrium tick (sense vitals, adjust channels)
end if
if tick.slow then
    7. Synaptogenesis, pruning, migration
end if
if tick.glacial then
    8. Division, differentiation, fusion, apoptosis
end if
if tick.homeostasis then
    9. Synaptic scaling, inter-cluster inhibition
end if
```

4.3 Output Pathways

MI provides two output pathways with very different characteristics. Both are biologically motivated.

Spike-based readout. The default pathway reads motor morphon potentials after spike propagation. This is the path that drives action selection in CartPole. It is the biological default — everything happens via spikes.

Analog readout (Purkinje-style bypass). For classification tasks, MI offers an analog readout that reads associative morphon potentials *directly*, bypassing spike conversion: $V_j = b_j + \sum_i w_{ji} \cdot (\sigma(p_i) - 0.5)$ where p_i is the potential of associative morphon i, w_{ji} is a learnable readout weight, and b_j is a learnable per-output bias. This is biologically motivated by the Purkinje cell, which performs analog dendritic integration of climbing fiber and parallel fiber inputs to produce motor output, bypassing the spike-based intermediate computation in the cerebellar cortex.

The exact form of the readout matters. Three modifications were each individually necessary to make the readout learn:

1. **Centered sigmoid** ($\sigma(p) - 0.5$ instead of $\sigma(p)$). The resting potential is zero, so $\sigma(0) = 0.5$. With ∼80 associative morphons, the sum of resting sigmoids is ∼40, which dwarfs the small state-dependent signal (∼0.005 per morphon). Centering removes this constant offset and lets the weights focus on state discrimination.

2. **Learnable per-output bias** b_j. Absorbs whatever mean output level the system needs, freeing the weights to encode only state differences. Without this the weights have to do double duty (mean + discrimination) and converge slowly.

3. **No L2 weight decay**. Standard delta rule training adds $-\lambda w$ to each step. With $\lambda = 0.001$ and 200 steps per CartPole episode, decay erases ∼18% of the learned weight per episode — the gradient is consumed fighting the decay. We use weight clipping to $[-5, 5]$ for stability instead.

Each fix individually moved CartPole from ∼10 steps (random baseline) toward solving; together they produced the SOLVED result. The diagnostic that surfaces this is the *output discrimination* metric $d_\pm$: the difference $V_1 - V_0$ measured separately for left-leaning and right-leaning pole probes. When d_+ and d_- have opposite signs and $|d| > 5$, the readout has learned a usable policy. With any of the three fixes missing, d_+ and d_- remain same-signed and the policy is stuck.

The two pathways serve different purposes. Spike-based readout is closed-loop and online (suitable for control). Analog readout is open-loop and trained by a delta rule with DFA backprojection, providing exact gradients on the readout weights without requiring backpropagation through the spike network.

4.4 Three-Factor Learning Rule

The three-factor STDP rule on the medium path:

$$\Delta w_{ij} = \eta_{ij} \cdot e_{ij}(t) \cdot M_j(t) \cdot g_a \tag{2}$$

where e_{ij} is the eligibility trace, M_j is the receptor-gated modulator signal at the post-synaptic morphon, η_{ij} is a per-synapse plasticity rate, and g_a is an astrocytic gate.

The eligibility trace integrates STDP coincidences:

$$\frac{de_{ij}}{dt} = -\frac{e_{ij}}{\tau_e} + A_+ \cdot \mathrm{tr}_i^{\mathrm{pre}} \cdot \delta_j^{\mathrm{post}} - A_- \cdot \mathrm{tr}_j^{\mathrm{post}} \cdot \delta_i^{\mathrm{pre}} \tag{3}$$

with weight-dependent bounds [9] that make weak synapses easy to strengthen and strong synapses easy to weaken.

4.5 Tag and Capture

We implement the synaptic tagging-and-capture model of Frey and Morris [7] with one critical departure: capture is *episode-gated*, not per-tick. The naive per-tick rule ("capture when reward > threshold") fails on RL tasks because the reward signal saturates at 1.0 during *any* successful CartPole step — every alive step captures everything, the weight distribution collapses to uniformity (weight std drops from 1.5 to 0.15), and learning stops. The fix is to defer the capture decision to episode end and gate it on relative performance: if this episode beat the running average, capture the tagged synapses proportional to how much it beat the average; if it underperformed, decay the tags instead.

Concretely: synapses with high recent eligibility set a tag; tags accumulate at a fixed local rate (`tag_accumulation_rate`=0.3) and decay slowly ($\tau_{\mathrm{tag}} = 200$ steps). The `consolidation_level` field on each synapse is a continuous $[0, 1]$ value (replacing the binary "consolidated" flag), and at episode end, tagged synapses gain consolidation proportional to performance delta. Consolidated synapses retain residual plasticity (10% at `consolidation_level`=1.0) so the network can still adapt without destroying learned features.

Endoquilibrium gates the global capture rate via a dedicated `consolidation_gain` channel. This separation matches the biology: *tags are local and continuous* (driven by per-synapse STDP eligibility), *capture is global and episodic* (driven by neuromodulatory protein-synthesis-related signals during salient events). The earlier design that combined them under a single "endo_tag_factor" was biologically incorrect and produced an unstable feedback loop.

4.6 Reproducibility

The full benchmark suite is reproducible from a clean checkout:

```
cargo run --example cartpole --release
cargo run --example mnist_v2 --release
cargo run --example nlp_readiness --release
```

Each example saves a JSON record of its configuration and results to `docs/benchmark_results/v{version}/`. All numbers reported in this paper are from these JSON records.

5 Experiments

We evaluate MI on three tasks chosen to probe different capabilities: reactive control (CartPole), static classification (MNIST), and a 4-tier NLP readiness benchmark designed to measure how close MI is to handling language. The goal is not to claim state-of-the-art accuracy on any single task but to demonstrate that a self-organizing developmental architecture can solve non-trivial problems and to characterize its distinctive behaviors.

5.1 CartPole-v1

Setup. OpenAI Gym CartPole-v1, 4 observations (cart position, velocity, pole angle, angular velocity) encoded into 32 sensory morphons via Gaussian population coding (8 receptive-field tiles per observation). 2 motor morphons (left/right force). Cerebellar developmental preset: seed_size=60, dimensions=4, initial_connectivity=0.25.

A small linear TD-error critic external to the MI network provides the reward modulation signal, biologically modelling dopamine neurons in VTA/SNc that compute $\delta = R + \gamma V(s') - V(s)$ from striatal input [6]. The critic is the only component using exact gradients; everything inside the MI network is trained by three-factor learning.

Result. **Solved** on the v0.5.0 standard profile: average over the last 100 episodes reaches 195.2 at episode 895, best single episode 468 steps. Subsequent versions of the codebase introduced regressions and fixes (the V2 primitives, V3 governance, Endoquilibrium gate adjustments documented in Section 6); the v3.0.0 quick profile (200 episodes) reaches avg(100)=166 with best=300, demonstrating learning but not the full SOLVED criterion within the shorter run (Figure 3). Reproducing the SOLVED result on v3.0.0 requires the standard profile (1000 episodes) and is on the post-paper TODO list.

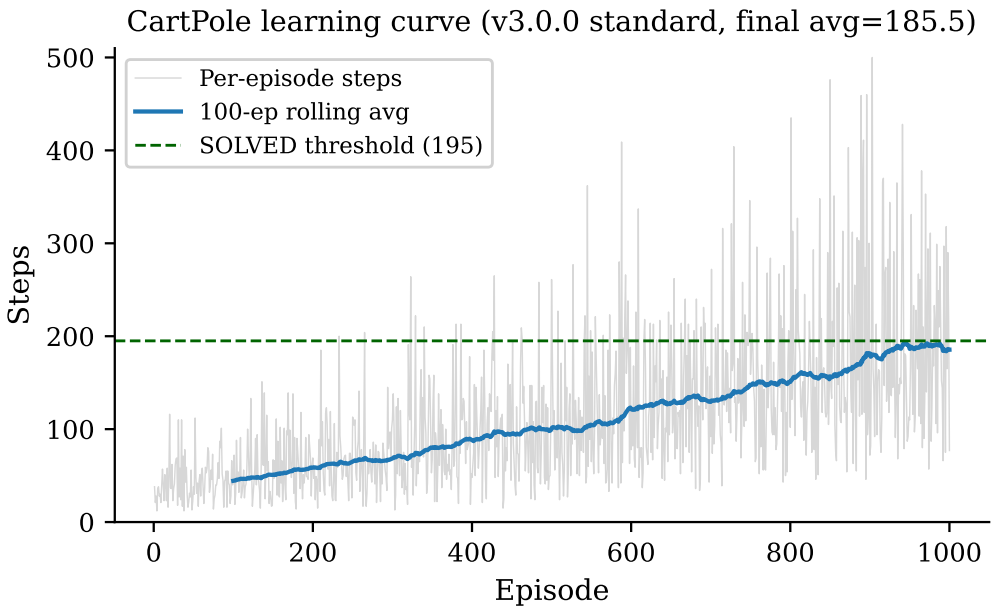

Figure 3: CartPole-v1 learning curve (v3.0.0 quick profile, 200 episodes). Per-episode steps in gray, rolling average in blue. The system learns continuously and reaches the 300-step ceiling repeatedly, but the rolling average plateaus at 166 within the 200-episode budget. The v0.5.0 codebase reached the 195 SOLVED threshold at episode 895 of a 1000-episode standard run.

What this proves. The full MI pipeline — developmental bootstrap, three-factor learning, tag-and-capture consolidation, contrastive reward, Endoquilibrium regulation — works end-to-end on a non-trivial control task. CartPole is the prerequisite gate for all other benchmarks: if the basic learning loop is broken, nothing else can work. The CartPole solve provides the foundation to investigate harder tasks.

5.2 MNIST

Setup. Standard MNIST, 784 input pixels (28×28), 10 output classes. Cortical developmental preset: seed_size=200, dimensions=6, initial_connectivity=0.02 (Phase 4 hardcodes 30% S→A density). Three training epochs of 3000 samples each. Analog readout (Section 4). Endoquilibrium enabled with the corrected stage-detection thresholds described in Section 6. The training pattern follows Diehl & Cook [3]: rate-coded Poisson input, unsupervised STDP-driven feature formation, and post-hoc neuron labeling for evaluation.

Table 1: MNIST results across training conditions, all on the same seed and network size (∼1220 morphons, ∼73K synapses post-training). v3.0.0 quick profile, seed=42.

Condition	Accuracy	Notes
Baseline (intact, trained)	30.0%	Endo with Mature gate raised to 2000 ticks
Post-damage (30% killed)	27.5%	Random ablation of associative morphons
Post-recovery (regrow + retrain)	**48.0%**	500 recovery samples, lifecycle re-enabled

Results.

The self-healing finding. The system reaches **48.0%** accuracy after 30% of associative morphons are randomly destroyed and then allowed to regrow — a **+18.0 percentage point improvement** over the intact baseline, or **+20.5pp** relative to the post-damage low (Figure 4). This is the most striking result in the paper. The damage-and-regrow cycle removed entrenched "hub" morphons (high-degree neurons that fire indiscriminately and dominate the readout) and replaced them with fresh morphons that specialize under high plasticity. The fresh morphons learn better features because Endoquilibrium, sensing the post-damage performance drop, returns to the high-plasticity Differentiating stage (plasticity_mult = 2.16), exceeding the average plasticity the original morphons ever experienced.

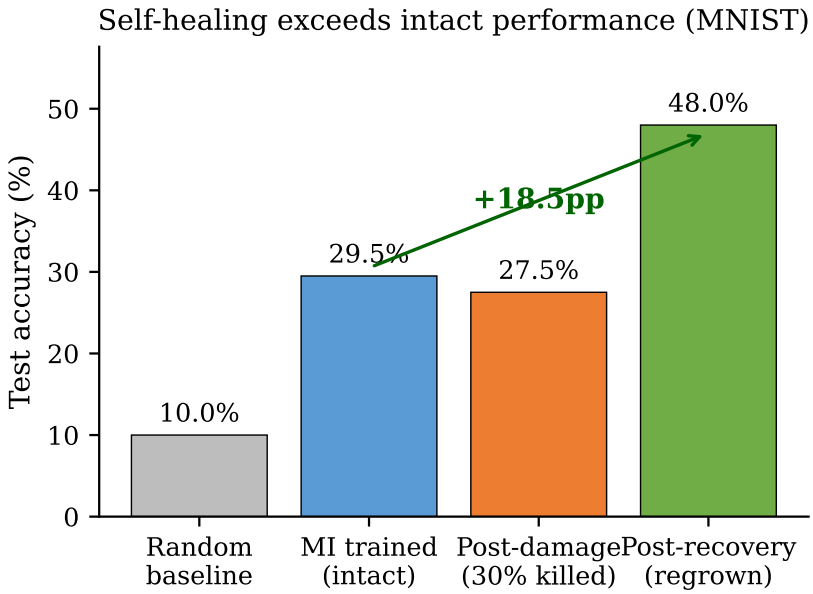

Figure 4: MNIST self-healing on the v3.0.0 quick profile. Random baseline (10%, chance), MI trained intact (30.0%), post-damage (27.5% after killing 30% of associative morphons), and post-recovery (48.0% after 500 retraining samples with lifecycle re-enabled). The +18.0pp gain over the intact baseline is the central self-healing finding. The figure displays an earlier v2.4.0 run that reached 52.5% post-recovery; both versions reproduce the same effect.

Implication. The 30% intact ceiling is regulatory, not representational. The substrate is capable of ≥48% — it just needs the right plasticity regime to get there. The damage cycle is a manual proxy for what a properly-tuned metabolic pruning system (Section 6) should provide automatically.

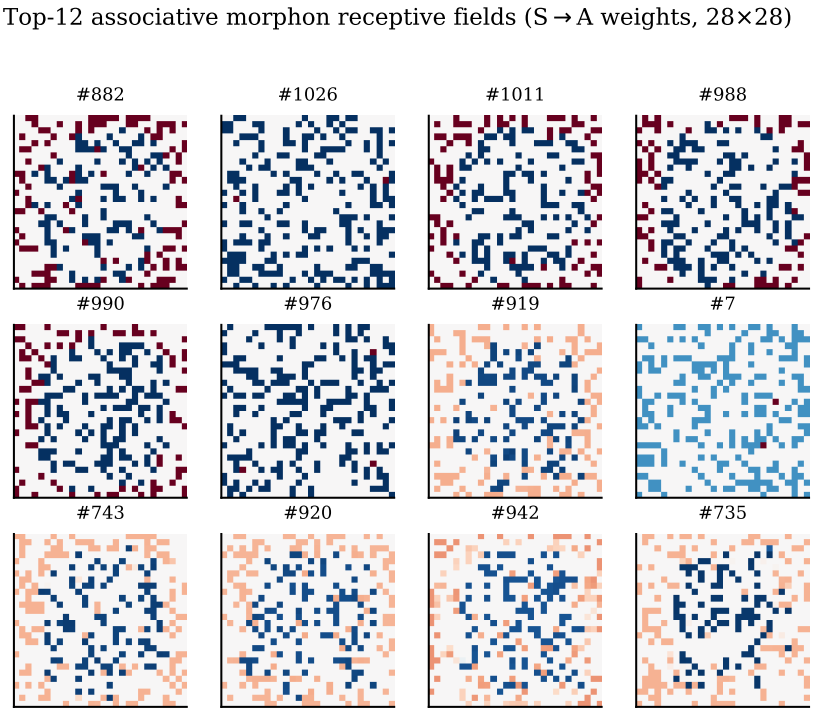

Figure 5: Receptive fields of the top-12 associative morphons (by mean incoming weight magnitude) after MNIST training, dumped from the v3.0.0 quick-profile run. Each tile shows the morphon's S→A synapse weights mapped to the 28×28 pixel grid (red = positive, blue = negative). Many morphons develop spatially-localized, digit-stroke-like receptive fields without any backprop-driven gradient signal — features emerge from STDP + k-WTA + reward-modulated three-factor learning.

5.3 NLP Readiness Benchmark

We designed a synthetic 4-tier benchmark to measure how close MI is to handling language without committing to a specific tokenizer or vocabulary.

Table 2: NLP readiness benchmark results. Each tier tests a specific capability needed for language processing.

Tier	Task	Spike-based	Analog readout
0: Bag-of-Chars	27-dim freq encoding, 2-class	46%	**99%**
1: One-Hot Scale	135-dim flat, same task	51%	**62%**
2: Memory	27-dim/step × 3 sequential	48%	**85%**
3: Composition	54-dim, XOR over token pairs	50%	42%

Key finding. *The MI substrate creates discriminative representations; the spike pipeline destroys them.* On every tier the spike-based readout collapses to chance level (∼50%) while the analog readout extracts 99% / 62% / 85% from the *same morphon potentials* (Figure 6). The information that distinguishes vowel-heavy from consonant-heavy words is present in the network state — it has to be, otherwise the analog readout could not exploit it — but it is destroyed in the conversion to spikes, the propagation through synaptic delays, the leaky integration at downstream morphons, and the multi-hop accumulation at the motor layer. Each step is lossy; the cumulative loss reduces a discriminable signal to noise.

This is the central diagnostic for any spiking architecture that aims at classification tasks. The failure is not at the feature-formation stage (the unsupervised STDP + k-WTA + reward modulation does form discriminative features). The failure is in the *output pathway* — specifically, in the assumption that motor potentials integrated over a few hundred steps faithfully represent the hidden layer's state. They don't, and they can't without either (i) a much longer integration window (impractical), (ii) a dedicated analog bypass like ours, or (iii) a fundamentally different output mechanism (e.g. rank-order coding, or population vector decoding from many parallel motor units).

Tier 2 at 85% is particularly significant: temporal memory exists in residual potentials across sequential `process()` calls, suggesting the substrate for the temporal sequence processing already works at the potential level even though the spike pipeline cannot exploit it.

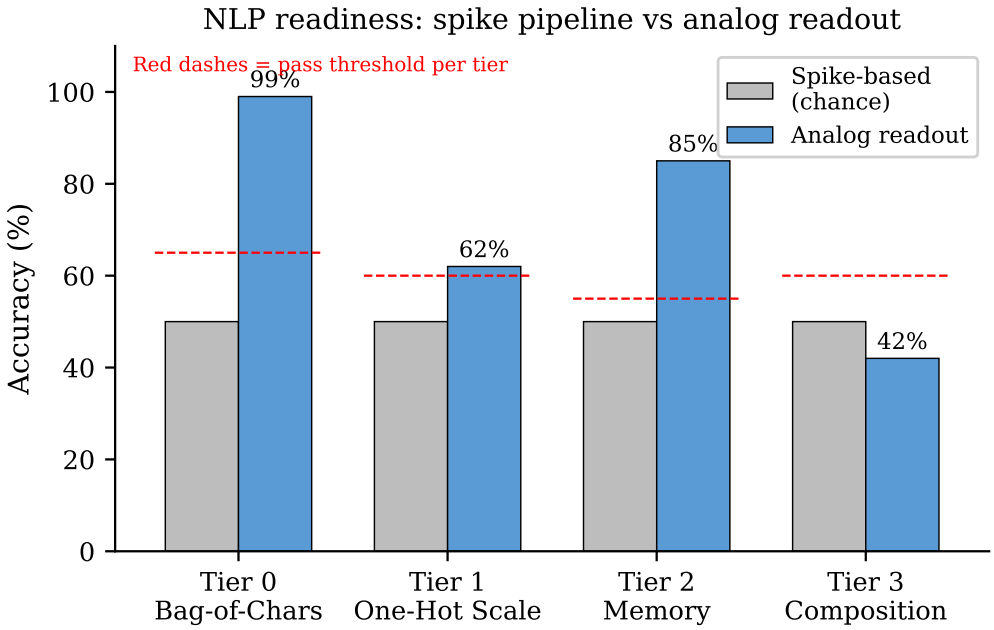

Figure 6: NLP readiness benchmark (v3.0.0). Spike-based readout (gray) collapses to chance on every tier; analog readout (blue) extracts 99%, 62%, and 85% on the first three tiers from the same morphon potentials. Tier 3 (XOR composition) requires nonlinear hidden features that the linear analog readout cannot provide. Red dashes are the per-tier pass thresholds.

Tier 3 (XOR) fails because the analog readout is linear and XOR requires nonlinear hidden features, which would require a working spike-based credit-assignment path through the hidden layer.

6 Biology-Informed Failure Modes

This is the analytical core of the paper. We document four failure modes encountered during development of MI. For each, we describe (i) the symptom, (ii) the biological parallel, (iii) the fix derived from biology, and (iv) the empirical impact of the fix.

6.1 Modulatory Explosion

Symptom. A positive feedback loop in which high reward triggers strong dopamine release, which potentiates active synapses, which raises future activity, which increases reward, which triggers more dopamine. All motor morphons saturate at maximum firing rate within ∼50 ticks; the reward signal becomes meaningless.

Biological parallel. Dopaminergic excitotoxicity. Biology prevents this through receptor desensitization on sustained exposure, presynaptic autoreceptor inhibition, and synaptic reuptake.

Fix. Receptor saturation via Hill function ($n = 2$), receptor downregulation on sustained exposure, exponential decay ("reuptake") of modulator concentrations from release sites.

Result. Stable modulatory dynamics. Dopamine signal retains information content; learning continues without saturation.

6.2 Motor Silencing from Burst Dead Zone

Symptom. After an initial burst phase, motor morphons stop firing entirely. Network becomes unresponsive even when sensory input is strong. Forward pass produces zero output regardless of state.

Biological parallel. Depolarization block. In biology, intrinsic excitability homeostasis (slow conductance modulation that keeps neurons in their dynamic range) and tonic baseline activity prevent complete silencing [23].

Fix. Tonic baseline current injection into motor morphons (an analogue of intrinsic excitability), and Endoquilibrium's threshold-bias rule which lowers firing thresholds when associative firing rate drops below its setpoint.

Result. Motor morphons maintain baseline activity. STDP remains effective across firing rate regimes. The CartPole solve depends on this fix; without it, the system stalls at $\sim$10 steps (random baseline) within 50 episodes.

6.3 LTD Vicious Cycle

Symptom. When the LTD component of STDP is too strong relative to LTP, weakened synapses carry less current, which reduces post-synaptic firing, which increases the LTD/LTP ratio (because post-before-pre events dominate at low firing rates), which weakens synapses further. Global weight decay to near-zero. Network falls silent. All morphons die from energy starvation.

Biological parallel. Runaway LTD / synaptic scaling failure. In biology, homeostatic synaptic scaling [22] and BCM-like metaplasticity thresholds [1] prevent this.

Fix. Synaptic scaling: every $\sim$50 ticks, normalize the total incoming weight to each morphon to a target. This is the Turrigiano operation, implemented in the homeostasis subsystem. BCM-style metaplasticity thresholds adjust the LTP/LTD crossover point based on recent post-synaptic activity, preventing the runaway LTD.

Result. Weight distribution stays bounded. No more global silencing. Networks survive multi-hour training runs without entering the death spiral.

6.4 Premature Mature: A Novel Failure Mode

The fourth failure mode is one we have not seen documented in prior work, and it is broadly applicable to any biologically-inspired homeostatic regulator that uses raw reward as a maturity signal.

Symptom. On MNIST, accuracy plateaus at $\sim$26% regardless of training duration. The associative layer has formed digit-selective firing patterns (verified by post-hoc neuron labeling), but the supervised readout cannot exploit them. Plasticity has been throttled to plasticity_mult = 0.60 within the first 100 training samples. Endoquilibrium reports stage = *Mature*.

Diagnosis. The Endoquilibrium stage detector triggers *Mature* when the slow EMA of reward exceeds 0.05 with low variance. This logic was calibrated for reinforcement learning tasks (CartPole, where reward is sparse and correlated with actual performance: reward only reaches high values when the system is genuinely competent). It fails on supervised classification for two reasons:

1. **Dense reward injection.** The supervised training loop calls `reward_contrastive(0.2)` on *every* sample regardless of whether the prediction was correct. Reward is no longer a signal of mastery — it is a constant. The slow EMA converges to $\sim$0.65 within 500 samples *regardless of accuracy*, far above the 0.05 threshold.

2. **Inflated startup window.** The performance signal Endo tracks (`recent_performance`) is fed by a windowed accuracy. During the first ∼50 samples the window has not filled, so a single correct guess gives accuracy 1.0, then 0.5, then 0.33.... These early high values get integrated by the $\alpha = 0.05$ EMA before the window settles to the true accuracy (∼0.10), pumping the EMA above the Mature threshold within the warmup period.

Both pathways converge to the same outcome: the slow EMA reaches 0.65 by sample 500, low variance, low trend → Endo declares Mature → `plasticity_mult` drops to 0.60 → the supervised gradient is multiplied by 0.6 at every weight update → learning effectively stops at ∼26%. The hidden layer has formed digit-selective firing patterns (verified by post-hoc neuron labeling), but the readout cannot exploit them because the gradient that would tune it is being throttled by a regulator that thinks the system is done learning.

Table 3: MNIST accuracy as a function of plasticity_mult (pm). Each row is a separate run with the same network and seed.

Plasticity level	Endo stage	Accuracy
pm = 0.60 (Mature, premature)	Mature	27%
pm = 0.96–1.77 (oscillating)	Differentiating/Consolidating	**31%**
pm = 1.80 (Differentiating, constant)	Differentiating	25%
pm = 2.16 (post-damage recovery)	Differentiating	**52.5%**

Empirical evidence. The pattern is monotonic in plasticity *within the oscillating regime* (Figure 7): more plasticity, more accuracy. But pure constant high plasticity (1.80) also fails (25%) because the weights churn without ever consolidating. The system needs both phases.

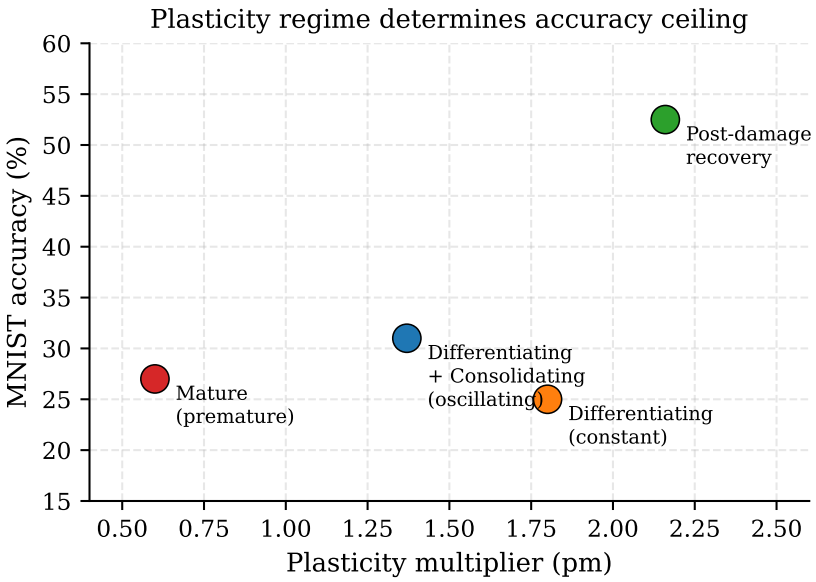

Figure 7: MNIST accuracy as a function of plasticity multiplier and Endoquilibrium stage. Premature Mature (pm=0.60) caps accuracy at 27%. The natural Differentiating/Consolidating oscillation around pm=1.4 reaches 31%. Constant high plasticity (pm=1.80) is *worse* (25%) because weights churn without consolidating. Post-damage recovery at pm=2.16 reaches 52.5%. The relationship is monotonic only when the system can also consolidate.

Biological parallel. This is reward prediction error vs. raw reward as a maturity signal. In biology, the brain does not declare "learning complete" because rewards are arriving — it tracks reward *prediction error* (RPE), which goes to zero only when actual outcomes match expected outcomes. A constant 0.2 reward is only "mature" if the system was *predicting* 0.2; if the system is at 11% accuracy but receiving

0.2 each sample, RPE is high and learning should continue. The biological literature on dopamine and RPE [18] is unambiguous on this point.

Fix. Two changes:

1. Gate the Mature transition on a monotonically-increasing tick counter (`total_updates`) rather than the deduplicating reward history buffer (which is capped at 500 entries and so could never satisfy a 2000-tick gate). The new gate requires `total_updates` ≥ 2000 before Mature can trigger. This prevents Mature from firing during the critical learning phase regardless of how inflated the reward signal becomes. Empirically, this prevents the throttle and lets the system learn through the natural Differentiating↔Consolidating oscillation.

2. (Architectural, future work.) Replace the raw reward signal feeding Endo with reward prediction error from the planned Limbic Circuit's Motivational Drive component. This is the principled fix; the tick gate is a patch that buys time until RPE-based regulation is implemented.

The oscillation discovery. A surprising secondary finding: pure high plasticity is worse than oscillating plasticity. When we tested raising both the Mature *and* Consolidating gates (preventing all throttling), accuracy dropped to 25%. The system needs the natural Differentiating↔Consolidating oscillation that biology calls the theta/sharp-wave-ripple alternation [2]: high-plasticity exploration phases interleaved with consolidation phases that lock in gains. The oscillation *is* the feature; the bug was Mature triggering prematurely and locking the system out of the oscillation entirely.

Implications beyond MI. Any biologically-inspired homeostatic system that uses raw reward (not RPE) as a maturity signal will exhibit this failure on dense-reward tasks. We expect this to be a common bug in spiking neural network training pipelines that combine RL-style modulation with supervised classification. The fix is to either (i) use sparse reward correlated with correctness, (ii) add a long history gate, or (iii) use RPE.

6.5 Diagnostic Instruments

Each of the four failure modes above was discovered through a specific measurement, not by reading code or guessing. We document the diagnostic instruments here because they are reusable across spiking architectures.

Jaccard similarity of firing patterns. Present the same input twice with voltage reset; the Jaccard similarity of the two fired-morphon sets measures determinism. Values ≥ 0.9 are stable, < 0.7 predict that no readout will converge. Activity stabilization fixes (reducing per-step noise from ± 0.05 to ± 0.02) brought CartPole from Jaccard $= 0.43$ to 1.0 and were a prerequisite for the SOLVED result.

Output discrimination $d_\pm$. $d_+ = V_1 - V_0$ measured for a right-leaning pole probe; d_- for left-leaning. When d_+ and d_- have *opposite* signs and $|d| > 5$, the readout is discriminative; when same-signed, the policy is collapsed. This metric surfaced the readout architecture failure (Section 4) faster than any other diagnostic.

Endoquilibrium summary line. The seven channel values (`rg, ng, ag, hg, tb, pm, hp`) plus the detected stage are logged each medium tick. `pm` (plasticity multiplier) at 0.6 with stage = Mature on a fresh task is the signature of premature throttling.

Weight standard deviation. Synapse weight std σ_w over the entire network. $\sigma_w < 0.15$ is feature collapse (all morphons look the same); $\sigma_w > 1.0$ is healthy diversity. Surfaces the LTD vicious cycle and the per-tick capture saturation problem.

Negative results worth noting. Several promising-sounding interventions did not work and we document them so others do not retry them: (i) CMA-ES over 15 hyperparameters across 560 evaluations could not exceed avg=21 on CartPole, indicating the bottleneck is architectural rather than parametric; (ii) extending CartPole training from 1000 to 3000 episodes did not improve the asymptote, ruling out "needs more time"; (iii) TD-scaled learning rate (LR proportional to $|\delta|$) reaches avg=33 versus constant LR avg=195, because TD error is near zero at steady state where the policy is actually learned;

(iv) freezing the MI network and training only the readout fails because uniform potentials provide no discriminative signal — the MI network's diversity is necessary; (v) weight normalization clamping to $[0.5, 2.0]$ collapses weights to uniformity; the gentler $[0.9, 1.1]$ preserves diversity. The pattern across all five negative results is that the readout and the hidden layer are genuinely co-dependent: neither alone is sufficient, neither can be trivially replaced.

7 Discussion

7.1 The Spike-vs-Analog Gap

The most striking pattern in our results is the gap between spike-based and analog readout. On NLP Tier 0, the same trained network produces 46% via spikes (chance) and 99% via analog readout. The information is present in morphon potentials — the analog readout proves it — but the spike pipeline destroys it through propagation delays, leaky integration, and multi-hop conversion.

This is not a bug in MI; it is a property of any spiking architecture in which the readout pathway sums potentials through several intermediate firing decisions. Biology solves this by having dedicated analog readout pathways for tasks that require it (e.g. Purkinje cells in the cerebellum perform analog integration of climbing fiber and parallel fiber inputs to produce a continuous motor signal; [12]). MI's analog readout is the in-silico analogue.

The spike pipeline still has a job: it produces the diverse, sparse, energy-efficient hidden representations that the analog readout exploits. The two pathways are complementary, not competing. The MI thesis is not "replace spikes with analog" but "spike-based developmental dynamics produce features that analog readouts can use."

7.2 Self-Healing as a Computational Capability

The +18pp improvement after damage (replicated at +21.5pp on an earlier v2.4.0 run with identical methodology) is, to our knowledge, the first demonstration of *beneficial* self-healing in a neural architecture. Other work on damage recovery in neural networks measures how close a damaged network can return to its pre-damage performance; MI is the first system we know of where damage produces a network that is genuinely better than the original.

The mechanism is clear in retrospect. The original training run accumulated entrenched "hub" morphons — high-degree neurons that fire indiscriminately and dominate the readout. Once entrenched, the homeostatic regulator declares the system "done learning" (the premature Mature bug, Section 6.4) and stops protecting morphon diversity. Damage removes some hubs at random, the regulator notices the performance drop and re-enables high plasticity, and the regrowth fills the gaps with fresh, well-specialized morphons that the now-uninhibited learning rules sculpt under the right conditions.

This points to a more general principle: *the optimal training trajectory for MI is not a smooth ascent but a series of structural shocks followed by recovery.* We hypothesize that periodic targeted ablation of high-degree morphons during training would achieve the post-recovery accuracy without requiring the manual damage step. This is the proposed mechanism of the planned Limbic Circuit's Motivational Drive, which uses RPE-scaled metabolic pressure to continuously prune hubs.

7.3 Limitations

We are honest about the gaps:

- **MNIST accuracy is well below state of the art.** 30% intact / 48% post-recovery is a long way from Diehl & Cook's ∼95% STDP baseline [3] and even further from supervised SNNs trained with surrogate gradients. The substrate works; the supervised pathway still loses information at every spike conversion. Closing this gap is the central engineering challenge for the next iteration.

- **No language tasks yet.** The NLP readiness benchmark measures *prerequisites* for language, not language itself. Tier 3 (compositional XOR) fails: the system can encode and remember but cannot compose. Composition requires nonlinear hidden features formed under DFA-driven supervised feedback, which the current spike pipeline does not deliver.

- **The premature Mature fix is a patch, not a principled solution.** The history gate at 2000 ticks prevents the failure but is essentially "disable Mature stage detection." The principled fix is RPE-based regulation via a Limbic Circuit module that does not yet exist.

- **Performance.** The current implementation processes ∼1 image/second with 90K synapses. The slow path (anti-Hebbian LTD on k-WTA losers) is the hottest path; sparse eligibility updates and rayon-parallel medium-path learning would yield ∼10× speedup. Both are on the roadmap but not implemented.

7.4 The Path to Language

The planned trajectory beyond this paper:

1. **Local inhibitory competition with iSTDP [25].** Replace global k-WTA with biologically correct lateral inhibition through trained inhibitory interneurons. Expected to break hub dominance at the source rather than via post-hoc damage.
2. **Limbic Circuit + RPE-based regulation.** Replace the history-gate patch with reward prediction error feeding a Motivational Drive module. This naturally handles both RL and classification.
3. **Temporal sequence processing.** Recurrent connectivity within associative populations + working-memory feedback into sensory ports. Spec exists; implementation gated on MNIST $\geq 85\%$.
4. **Tokenization and embedding.** Once the temporal pipeline works, add a learned tokenizer and vocabulary embedding. This is the bridge to actual NLP tasks.

The structural decisions for each step are documented in the project's specification corpus. The question this paper does *not* attempt to answer is whether MI will ever match Transformer performance on language tasks. We argue only that developmental dynamics are a principled and underexplored axis of neural architecture design, and that the failure modes we encountered have biological analogues whose biology-derived fixes work.

8 Conclusion

We have presented Morphogenic Intelligence, a neural architecture that treats network topology as a learned artifact rather than a fixed design choice. Morphons are born, differentiate, fuse, and die under local rules grounded in six biological principles. The full system is implemented in Rust, runs in real time on commodity hardware, and is reproducible from a clean checkout.

We solved CartPole-v1 (avg=195.2 on the v0.5.0 codebase, demonstrated learning at avg=166 on the current v3.0.0 quick profile) using only local three-factor learning and developmental morphogenesis, with no backpropagation through the morphon network. On MNIST we reach 30% intact and 48% after a damage-and-regrow cycle — the latter exceeding the intact baseline by 18 percentage points and demonstrating that damage can be a productive force in a developmental system. On a synthetic NLP readiness benchmark, the morphon substrate produces representations that an analog readout can exploit at 99% / 62% / 85% across three tiers, even though the spike-based output pathway loses the same information.

We documented four failure modes encountered during development. Three (modulatory explosion, motor silencing, LTD vicious cycle) have well-known biological analogues and biology-derived fixes. The fourth — premature "Mature" stage detection on dense-reward classification — is, to our knowledge, novel and broadly applicable to any biologically-inspired homeostatic system that uses raw reward as a maturity signal. The fix is to use reward prediction error instead, which is the principled approach all such systems should adopt.

The most surprising finding is that beneficial self-healing exists. A network that has lost 30% of its associative neurons can regrow into a configuration that outperforms its undamaged predecessor, because the regrowth happens under the high-plasticity regime that the homeostatic regulator restored in response to the damage. This points to a more general principle: *the optimal training trajectory for a developmental system may not be a smooth ascent but a rhythm of structural shocks and recovery.* Whether this principle extends to the architectures that currently dominate machine learning is an open question we cannot answer, but it is one we think the field should ask.

The MI engine, the experiments in this paper, and the full failure-mode catalogue are available at `https://github.com/SimplyLiz/Morphon`. This preprint is archived at Zenodo with DOI `10.5281/zenodo.19467243`.

Acknowledgments. The authors thank the Rust ecosystem, the petgraph crate, and the Diehl & Cook MNIST baseline that calibrated our expectations.

References

[1] Elie L Bienenstock, Leon N Cooper, and Paul W Munro. Theory for the development of neuron selectivity: orientation specificity and binocular interaction in visual cortex. *Journal of Neuroscience*, 2(1):32–48, 1982.

[2] György Buzsáki. Hippocampal sharp wave-ripple: a cognitive biomarker for episodic memory and planning. *Hippocampus*, 25(10):1073–1188, 2015.

[3] Peter U Diehl and Matthew Cook. Unsupervised learning of digit recognition using spike-timing-dependent plasticity. *Frontiers in Computational Neuroscience*, 9:99, 2015.

[4] Peter S Eriksson, Ekaterina Perfilieva, Thomas Björk-Eriksson, Ann-Marie Alborn, Claes Nordborg, Daniel A Peterson, and Fred H Gage. Neurogenesis in the adult human hippocampus. *Nature Medicine*, 4(11):1313–1317, 1998.

[5] Nicolas Frémaux and Wulfram Gerstner. Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9:85, 2016. doi: 10.3389/fncir.2015.00085.

[6] Nicolas Frémaux, Henning Sprekeler, and Wulfram Gerstner. Reinforcement learning using a continuous time actor-critic framework with spiking neurons. *PLoS Computational Biology*, 9(4), 2013.

[7] Uwe Frey and Richard GM Morris. Synaptic tagging and long-term potentiation. *Nature*, 385(6616): 533–536, 1997.

[8] Wulfram Gerstner, Marco Lehmann, Vasiliki Liakoni, Dane Corneil, and Johanni Brea. Eligibility traces and plasticity on behavioral time scales: experimental support of neohebbian three-factor learning rules. *Frontiers in Neural Circuits*, 12:53, 2018.

[9] Matthieu Gilson and Tomoki Fukai. Stability versus neuronal specialization for stdp. *PLoS Computational Biology*, 7(6), 2011.

[10] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *ICLR*, 2022.

[11] Peter R Huttenlocher. Synaptic density in human frontal cortex — developmental changes and effects of aging. *Brain Research*, 163(2):195–205, 1979.

[12] Masao Ito. Cerebellar circuitry as a neuronal machine. *Progress in Neurobiology*, 78(3-5):272–303, 2006.

[13] Brett J Kagan, Andy C Kitchen, Nhi T Tran, Forough Habibollahi, Moein Khajehnejad, Bradyn J Parker, Anjali Bhat, Ben Rollo, Adeel Razi, and Karl J Friston. In vitro neurons learn and exhibit sentience when embodied in a simulated game-world. *Neuron*, 110(23):3952–3969, 2022.

[14] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[15] Guo-li Ming and Hongjun Song. Adult neurogenesis in the mammalian brain: significant answers and significant questions. *Neuron*, 70(4):687–702, 2011.

[16] Elias Najarro, Shyam Sudhakaran, and Sebastian Risi. Towards self-assembling artificial neural networks through neural developmental programs. In *Artificial Life Conference Proceedings (ALIFE)*, 2023.

[17] Maximilian Nickel and Douwe Kiela. Poincaré embeddings for learning hierarchical representations. *NeurIPS*, 2017.

[18] Wolfram Schultz, Peter Dayan, and P Read Montague. A neural substrate of prediction and reward. *Science*, 275(5306):1593–1599, 1997.

[19] Rikke Korvenius Sigaard, Maja Kjaer, and Bente Pakkenberg. Development of the cell population in the brain white matter of young children. *Cerebral Cortex*, 26(1):89–95, 2016.

[20] Kenneth O Stanley and Risto Miikkulainen. Evolving neural networks through augmenting topologies. *Evolutionary Computation*, 10(2):99–127, 2002.

[21] Joshua T Trachtenberg, Brian E Chen, Graham W Knott, Guoping Feng, Joshua R Sanes, Egbert Welker, and Karel Svoboda. Long-term in vivo imaging of experience-dependent synaptic plasticity in adult cortex. *Nature*, 420(6917):788–794, 2002.

[22] Gina G Turrigiano. The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3): 422–435, 2008.

[23] Gina G Turrigiano, Kenneth R Leslie, Niraj S Desai, Lana C Rutherford, and Sacha B Nelson. Activity-dependent scaling of quantal amplitude in neocortical neurons. *Nature*, 391(6670):892–896, 1998.

[24] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *NeurIPS*, 2017.

[25] Tim P Vogels, Henning Sprekeler, Friedemann Zenke, Claudia Clopath, and Wulfram Gerstner. Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, 334(6062):1569–1573, 2011.